

TEAM VIRUS PROJECT REPORT

Course: Computational Thinking and Programming (2025CSET100)

Project Title: Library Management System

Batch: 35

Team Number: _____

Team Name: Team Virus

Date: November 2025

Team Members:

- Ritika Sharma (Team Leader)
 - Sparsh Sinha
 - Shauryaditya Tiwari
-

LIBRARY MANAGEMENT SYSTEM – PROJECT REPORT

1. Introduction

The Library Management System is a Python-based application designed to simplify and automate essential library tasks. It uses a Tkinter GUI and stores all book records in CSV format to keep data safe and organized. This system is suitable for schools, colleges, and small libraries that need a basic digital solution.

2. Problem Statement

Manual library management is slow, error-prone, and hard to maintain. Searching books, tracking issued or returned books, and updating registers take time and often lead to mistakes. A digital system is required to make these processes faster and more accurate.

3. Objectives

- To automate basic library operations
- To create an easy and user-friendly GUI using Tkinter
- To store and manage records using CSV
- To reduce manual errors and save time
- To apply programming and problem-solving skills

4. Proposed Solution

The solution is a GUI-based Library Management System built using Python. It allows users to add books, view books, search books, issue books, return books, and delete unwanted records.

All updates are saved permanently into a CSV file, ensuring accuracy and reliability.

5. System Features

- Add Book – Add new books with title and author
- View Books – Display all books in a table (TreeView)
- Search Book – Search using Book ID or title
- Issue Book – Issue a book with borrower name and date
- Return Book – Mark the book as returned
- Delete Book – Remove a book record permanently
- CSV Storage – Secure permanent data storage

6. System Workflow

User → GUI (Tkinter) → Select Operation → Python Functions Run → CSV File Updates → Result Shown on GUI

7. Technologies Used

- Python 3
- Tkinter GUI Library
- CSV File Handling
- TreeView Widget
- Datetime Module
- Basic Functions and Logic

8. Results

The system works successfully and performs all operations smoothly. It makes book management easier, faster, and error-free. The GUI is simple to understand and ensures proper data handling.

9. Conclusion

The Library Management System is a reliable and simple project that automates essential library activities. It demonstrates Python GUI development, file handling, and logic building.

Future enhancements may include login system, fine calculator, or cloud backup.

10. Individual Contributions

- Sparsh Sinha – GUI Design, PPT
- Ritika Sharma – Resources and Content
- Shauryaditya Tiwari – Report Writing

END OF PROJECT REPORT